

# **Fomoco**

## Mathematics & Methodology Report

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*Professional Quantitative Documentation of Long-Term Investment Visualization Tool*

## **1 Mathematics Behind Fomoco**

Fomoco visualizes the long-term impact of regular monthly investing versus leaving money in a low-yield bank account. All formulas are implemented in `calculations.py` and rendered interactively in the Streamlit application.

### **1.1 Monthly Compounding Rate**

$$r_m = \frac{r}{12 \times 100} = \frac{r}{1200} \quad (1)$$

where  $r$  is the annual expected return in percent.

### **1.2 Future Value of Annuity (Monthly Contributions)**

For monthly investment  $P$  over  $n$  months at monthly rate  $r_m$ :

$$FV(n) = P \times \frac{(1 + r_m)^n - 1}{r_m} \quad (\text{when } r_m \neq 0) \quad (2)$$

Special case (zero interest):

$$FV(n) = P \times \frac{n}{12} \quad (3)$$

The function returns only **end-of-year** values:

$$\text{years\_list} = [1, 2, \dots, T], \quad FV_{\text{yearly}} = FV[11 :: 12]$$

### 1.3 Risk Bands

For a chosen basket with expected return  $r$  and volatility  $\sigma$ :

$$\begin{aligned} \text{lower\_rate} &= \max(r - 0.8\sigma, -2.0) \\ \text{expected\_rate} &= r \\ \text{upper\_rate} &= r + \sigma \end{aligned}$$

The three curves are:

$$\begin{aligned} \text{lower}(t) &= FV_t(P, \text{lower\_rate}) \\ \text{expected}(t) &= FV_t(P, \text{expected\_rate}) \\ \text{upper}(t) &= FV_t(P, \text{upper\_rate}) \end{aligned}$$

Basket parameters:

| Basket               | Expected Return $r$ | Volatility $\sigma$ |
|----------------------|---------------------|---------------------|
| Conservative (bonds) | 5.0%                | 4.0%                |
| Balanced (60/40)     | 8.0%                | 9.0%                |
| Aggressive (stocks)  | 12.0%               | 18.0%               |

### 1.4 Bank Account Comparison

$$\text{bank\_FV}(t) = FV_t(P, 0.25) \tag{4}$$

### 1.5 Final Values and Opportunity Cost

$$\begin{aligned} \text{Final}_{\text{investment}} &= \text{expected}[T] \\ \text{Final}_{\text{bank}} &= \text{bank\_FV}[T] \\ \Delta &= \text{Final}_{\text{investment}} - \text{Final}_{\text{bank}} \end{aligned}$$

## 2 Sources & Industry Context (Fully Validated May 2026)

All return and volatility assumptions are illustrative educational benchmarks. They were independently verified against live industry reports and historical data as of May 2026.

### 2.1 Assumptions vs Industry Benchmarks

Table 1: Comparison of Fomoco Assumptions vs. Industry Sources (Expected Annual Return / Volatility)

| Source                | Conservative (Bonds) | Balanced (60/40)   | Aggressive (Stocks)  |
|-----------------------|----------------------|--------------------|----------------------|
| <b>Fomoco</b>         | <b>5.0% / 4.0%</b>   | <b>8.0% / 9.0%</b> | <b>12.0% / 18.0%</b> |
| J.P. Morgan 2026      | 4.6% / ~5%           | 6.4% / ~11%        | 7.0% / ~16.5%        |
| Verus 2026 CMA        | 4.8% / 4.9%          | 6.3% / 11.0%       | 7.3% / 16.6%         |
| Hist. Avg (1985–2026) | 5–6% / ~5%           | 7.36% / 9.03%      | 10–12% / 15–20%      |

### 2.2 Direct Source References (All Links Verified & Working)

1. **J.P. Morgan Asset Management — 2026 Long-Term Capital Market Assumptions** (30th edition)

Full report PDF: <https://am.jpmorgan.com/content/dam/jpm-am-aem/americas/us/en/institutional/insights/lcma-full-report.pdf>

(Confirmed: U.S. Bonds 4.6%, Global Equities 7.0%, 60/40 6.4%)

2. **Verus Investments — 2026 Capital Market Assumptions**

Full PDF: <https://www.ipopif.org/Resources/a96343a6-b25e-47d2-b857-99d4e7bea0fe/Verus%202026-Capital-Market-Assumptions.pdf>

(Confirmed: Core Fixed Income 4.8% / 4.9% vol, Global Equity 7.3% / 16.6% vol)

3. **Historical Performance Analysis (60/40 Portfolio)**

Longtermtrends chart: <https://www.longtermtrends.com/60-40-stocks-bonds-portfolio/>

(Confirmed: 7.36% return / 9.03% volatility)

These figures are widely used in financial education platforms because they clearly illustrate the power of compounding while staying grounded in real industry data.